

Research Methodology

Writing SLR in Steps

Guide to Systematic Literature Review

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What is a Systematic Literature Review?

- An SLR is a comprehensive and methodical approach to reviewing literature in a specific field. It involves identifying, selecting, evaluating, and synthesizing all available research related to a specific research question.
- Systematic reviews are crucial for summarizing evidence, identifying gaps, and guiding future research.
- Compared to traditional literature reviews, SLRs are more structured and transparent, which helps minimize bias.

Benefits of Conducting a SLR:

- **Comprehensive Analysis:** Captures all relevant research on a topic.
- **Objective Approach:** Reduces bias through systematic methodology.
- **Reliable Evidence Base:** Enables robust conclusions and recommendations.
- **Identification of Gaps:** Highlights areas where research is lacking.



PICO Framework Timeline

- The **PICO framework** originated in the field of **evidence-based medicine (EBM)** as a way to structure clinical questions for research and systematic reviews.
- It was **introduced in 1995 by Scott Richardson and colleagues** to help clinicians and researchers break down complex questions into searchable components:
 - **P** – Patient, Problem, or Population
 - **I** – Intervention
 - **C** – Comparison (or Control)
 - **O** – Outcome
- Over time, PICO became a cornerstone of **Evidence-Based Practice (EBP)** and is widely used in systematic reviews, clinical guidelines, and research design.
- Variants like **PICOT** (adding Time) and **PICOS** (adding Study design) were later developed for more specialized contexts.
- Although it started in healthcare, scholars have argued that PICO can be adapted for **any scientific discipline**, including **Data Science and AI**, because its structure aligns with universal research components:
 - Problem → Research object
 - Intervention → Method or algorithm
 - Comparison → Alternative method or baseline
 - Outcome → Knowledge or performance metric.



PICO Framework AI/DS

- The **PICO framework** is traditionally used in evidence-based research (especially healthcare) to structure questions, but it can be adapted for quantitative topics in **Data Science (DS)** and **AI** to define clear research or experimental questions.
- Here's how you can apply **PICO** to a DS/AI context:
- **PICO Components for DS/AI**
- **P (Population / Problem)**
 - Define the dataset or domain you are studying.
 - Example: *Tweets from healthcare professionals, Customer reviews on e-commerce platforms, Medical imaging data for cancer detection.*
- **I (Intervention)**
 - The algorithm, model, or technique you are applying.
 - Example: *Transformer-based sentiment analysis, Random Forest for fraud detection, CNN for image classification.*
- **C (Comparison)**
 - The baseline or alternative method you compare against.
 - Example: *Traditional Bag-of-Words model, Logistic Regression, Previous state-of-the-art architecture.*
- **O (Outcome)**
 - The measurable result or metric you aim to improve.
 - Example: *Accuracy, F1-score, RMSE, AUC, Reduction in false positives, Improved interpretability.*



How PICO Shapes an AI Research Question

- Using sentiment analysis as an example:
- **P (Population / Problem)**
Customer reviews on e-commerce platforms
→ Defines the domain and dataset.
- **I (Intervention)**
Transformer-based model (e.g., BERT)
→ Specifies the technique you want to test.
- **C (Comparison)**
Traditional LSTM or Bag-of-Words approach
→ Sets a baseline for evaluation.
- **O (Outcome)**
Improved F1-score and accuracy
→ States the measurable goal.
- **Resulting Question:**
 - In sentiment analysis of customer reviews, does using a BERT-based model compared to an LSTM improve classification accuracy and F1-score?

How It Helps Find Gaps

- **Clarifies what's missing:** If most studies compare BERT vs. LSTM but not BERT vs. domain-specific transformers, that's a gap.
- **Highlights underexplored outcomes:** Maybe prior work focuses on accuracy but ignores interpretability or computational cost.
- **Reveals neglected populations:** For example, multilingual sentiment analysis might be less studied than English-only datasets.



Step 1: Define Your Research Question

- The first step in conducting a systematic literature review is formulating a clear, concise research question.
- An effective way to do this is by using frameworks like PICO (Population, Intervention, Comparison, Outcome) for **quantitative research** or SPIDER (Sample, Phenomenon of Interest, Design, Evaluation, Research type) for **qualitative research**.
- **How to Formulate a Good Question:**
 - Be specific and focused.
 - Align the question with your research objectives.
 - Ensure it addresses a gap in the existing literature.
- **Example:**
 - For a medical SLR: *"What is the efficacy of drug X compared to drug Y in reducing symptoms of condition Z in adults?"*

Step 2: Develop a Protocol

- A protocol outlines the methods and criteria you will use in your SLR.
- It ensures consistency and transparency throughout the review process.
- Your protocol should include:
- **Objective:** What are you trying to accomplish?
- **Eligibility Criteria:** Define inclusion and exclusion criteria.
- **Search Strategy:** Outline databases, keywords, and search strings.
- **Data Extraction Plan:** Describe how you will collect and analyze data.
- **Quality Assessment:** Plan how you will assess the quality of the studies.
- **Example Protocol Component:**
- **Eligibility Criteria:** Include peer-reviewed articles from 2015 onwards; exclude non-English publications.



Step 3: Conduct a Comprehensive Literature Search

- Your search strategy must be systematic and replicable. Use multiple **academic search engines** like PubMed, Web of Science, and Google Scholar.
- Consider grey literature to avoid publication bias. To ensure completeness, conduct pilot searches to refine your strategy.
- **Tips for an Effective Search:**
 - Use Boolean operators (AND, OR, NOT) to combine keywords.
 - Utilize advanced search features to refine your results.
 - Keep track of your search process for transparency.
 - Use synonyms and related terms to expand your search scope.



Step 4: Screen and Select Studies

- Screening involves reviewing the titles, abstracts, and full texts of the identified studies to determine their relevance.
- Follow these steps:
- **Initial Screening:** Exclude studies that clearly do not meet inclusion criteria.
- **Full-Text Review:** Evaluate the remaining studies for eligibility.
- **Data Extraction:** Extract key data such as author, year, methodology, and outcomes.
- **Screening Example:**
 - Initial screening identified 500 studies; after title and abstract screening, 150 were selected for full-text review.



Step 5: Assess the Quality of the Studies

- Assessing the quality of included studies is vital to ensure your review's credibility. Use standardized quality assessment tools such as:
- **Cochrane Risk of Bias Tool:** For randomized controlled trials.
- **Newcastle-Ottawa Scale:** For observational studies.
- **CASP Checklists:** For qualitative research.
- **Quality Assessment Tip:**
- Create a scoring system to rank the studies based on methodological soundness.

Step 6: Data Extraction and Management

- Create a structured data extraction form to collect consistent information.
- **Include:**
 - Study characteristics (author, year, country)
 - Population and sample size
 - Methodological details
 - Key findings and outcomes
- **Recommended Tools:**
 - Use Excel or specialized software like RevMan for data management.



Step 7: Data Synthesis

- Synthesize data to identify patterns and draw conclusions.
- Depending on your data type, choose between:
 - *Narrative Synthesis: For qualitative data, summarizing themes and insights.*
 - **Meta-Analysis:** For quantitative data, pooling results using statistical techniques.
- **Synthesis Example:**
 - Group studies based on intervention types and compare outcomes to identify trends.

Step 8: Interpret the Findings

- Interpretation should go beyond summarizing.
- Critically discuss:
- **Consistency:** Are the findings in agreement?
- **Heterogeneity:** Are there differences across studies?
- **Implications:** What do the results mean for your field of study?
- **Limitations:** Acknowledge any potential biases or gaps.



Step 9: Report Your Findings

- Follow reporting guidelines such as PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) to present your SLR clearly and systematically.
- Key components include:
- **Abstract:** Brief summary of the review.
- **Introduction:** Background and objectives.
- **Methods:** Detailed methodology and protocol.
- **Results:** Findings from the synthesis.
- **Discussion:** Interpretation and implications.
- **Conclusion:** Final thoughts and future research directions.



Step 10: Update Your Review

- Since knowledge is constantly evolving, periodically updating your systematic review ensures its relevance.
- Set a schedule to re-run searches and incorporate new findings.



Hands on Write SLR in Lab

- Step-by-step guide to conducting a Systematic Literature Review (SLR) on
 - **"NLP for Domain-Specific Sentiment Analysis"**

